

Product Specification

SKU	DMHSM0012VGNA
Vendor	DisplayModule
Version	V1.0.0

Version	data	Modification	Auditor
V1.0.0	Dec 13, 2024	First Create	Stan

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1.0 GENERAL DESCRIPTION

1.1 Introduction

The DMHSM0012VGNA silicon-based micro-display employs a fully digital driving solution, supporting 8-bit parallel data input. It is manufactured using a 0.18-micron CMOS process. The display's orientation, position, and built-in testing functions can be controlled and adjusted through a 2-wire serial programming interface. The interface voltage standard is 1.8V CMOS, making it suitable for a wide range of miniaturized, high-resolution near-eye display systems.

1.2 Features

- 0.18-micron CMOS process
- Utilizes active matrix driving technology
- SPI serial communication interfaces
- 8-bit parallel data input
- Supports row decoding random
- Multiple test patterns are built in
- RoHS Compliant

1.3 General Specification

Item	Contents
Screen Size	0.12"
Mono or RGB	Mono(Green)
Resolution	640(H)*480(V)
Effective Pixel	656(H)*496(V)
Pixel Size	3.75(H)*3.75(V) um
Display Area	2.49(H)*1.86(V) mm
Module Size	17.58(H)20.82(V)1.5(D) mm
Gray Level	Determined by driver
Interface	SPI/QSPI
PPI	6800
Brightness (nits)	5,000,000
Dynamic Range (Contrast Ratio)	100,000: 1
COM	0V-3V(determined by device characteristics)
Response Time	ns
Product Life (hour)	100,000

2.0 ELECTRICAL SPECIFICATIONS

Parameter	Symbol	Min.	Typ.	Max.	Unit
Digital Power Supply	V_{DD}	1.62	1.8	1.92	V
Common Cathode	V_{COM}		-1		V
Pixel Array Power Supply	V_{DDP}		1.8		V
Power Consumption	P_{total}			100	mW
Gate Driver High Voltage	V_{IH}	$0.7V_{DD}$		V_{DD}	V
Gate Driver Low Voltage	V_{IL}	0		$0.3V_{DD}$	V

3.0 Pin Definitions

3.1 Connector Pin Assignment

Pin Order	Symbol	Description
P1	VCC 1V8	1.8V Power Supply
1	SLA	I2C Address Selection
2	SCL	I2C Serial Communication Clock Line (When SCL is not in use, it must still be pulled up)
3	VCC 0V9	ASICS 0.9V Power Supply
4	RST	VGA RST
5	GND	Power Ground
6	GND	Power Ground
7	GND	Power Ground
8	IO1	SPI/QSPI Data Input/Output
9	SDA	I2C Serial Communication Data Line
10	SPI_MOSI_F/IO0	SPI/QSPI Data Input/Output
11	CS	Bit select signal
12	IO3	QSPI Data Input/Output
13	SPI_SCK_F/SCLK	Clock signal
14	IO2	QSPI Data Input/Output

P2	VCOM	Common Cathode
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3.2 Chip Pin Assignment

The top IO definitions of the chip are shown below. The top IO port numbers are ordered from left to right. The coordinate values are the differences between adjacent coordinates (referring to the spacing between the left top edges of two pads). Taking T2 as an example, the coordinate value of T2 is the difference between T2 and T1, and so on (with the left top edge of the pad chosen as the unified reference point).

The top IO definitions of the chip				
Pin Order	Symbol	Description	Note	Coordinates (μm)
T1	VDD	Digital 1.8V Power Supply	Typical : 1.8V	
T2	VCOM	Common Cathode	Typical : -1V*	145.1
T3	SIGNAL3[0:3]	Signal Port		275.3
T4				80
T5				80
T6				80
T7	VDD	Digital 1.8V Power Supply	Typical : 1.8V	80
T8	SIGNAL8	Signal Port		145.1
T9	SIGNAL9	Signal Port		80
T10	SIGNAL10	Signal Port		80
T11	VSS	Ground	Typical : 0V	80
T12	SIGNAL12[0:3]	Signal Port		145.1
T13				80
T14				80
T15				80
T16	VDD	Digital 1.8V Power Supply	Typical : 1.8V	80
T17	SCL_P	I2C Serial Communication Clock Line	Typical : 1.8V	145.1
T18	SDA_P	I2C Serial Communication Data Line	Typical : 1.8V	80
T19	SIGNAL19	Signal Port		80

T20	SLA1_P	I2C address		80
T21	SLA2_P			80
T22	SIGNAL22	Enable		80
T23	VSS	Ground	Typical : 0V	80
T24	SIGNAL24	Signal Port		145.1
T25	SIGNAL25	Signal Port		80
T26	VCOM	Common Cathode	Typical : -1V*	80
T27	VDD	Digital 1.8V Power Supply	Typical : 1.8V	275.3

Note: IO ports marked with * require reference to the turn-on voltage of μ LED devices.

The right side IO definitions of the chip are shown below, where the port numbers are ordered from top to bottom. Coordinate values represent the differences between adjacent coordinates (referring to the spacing between the top edges of two pads). For example, for R2, the coordinate value is the difference between R2 and R1, and so forth (with the top edge of the pad chosen as the unified reference point).

The right side IO definitions of the chip				
Pin Order	Symbol	Description	Note	Coordinates (μ m)
R1	VSS	Ground	Typical : 0V	
R2	VCOM	Common Cathode	Typical : -1V*	126.65
R3	VDD	Digital 1.8V Power Supply	Typical : 1.8V	390.6
R4	VCOM	Common Cathode	Typical : -1V*	390.6
R5	VDD	Digital 1.8V Power Supply	Typical : 1.8V	390.6
R6	VCOM	Common Cathode	Typical : -1V*	390.6
R7	VDD	Digital 1.8V Power Supply	Typical : 1.8V	390.6

Note: IO ports marked with * require reference to the turn-on voltage of OLED or LED devices.

The bottom IO definitions of the chip are shown below, where the port numbers are ordered from left to right. Coordinate values represent the differences between adjacent coordinates (referring to the spacing between the left top edges of two pads). For example, for B2, the coordinate value is the difference between B2 and B1, and so forth (with the left top edge of the pad chosen as the unified reference point).

The bottom IO definitions of the chip				
Pin Order	Symbol	Description	Note	Coordinates (μ m)
B1	VDD	Digital 1.8V Power Supply	Typical : 1.8V	

B2	VCOM	Common Cathode	Typical : -1V*	455.7
B3	VDD	Digital 1.8V Power Supply	Typical : 1.8V	520.8
B4	VCOM	Common Cathode	Typical : -1V*	520.8
B5	VDD	Digital 1.8V Power Supply	Typical : 1.8V	520.8
B6	VCOM	Common Cathode	Typical : -1V*	520.8

Note:IO ports marked with * require reference to the turn-on voltage of μ LED devices.

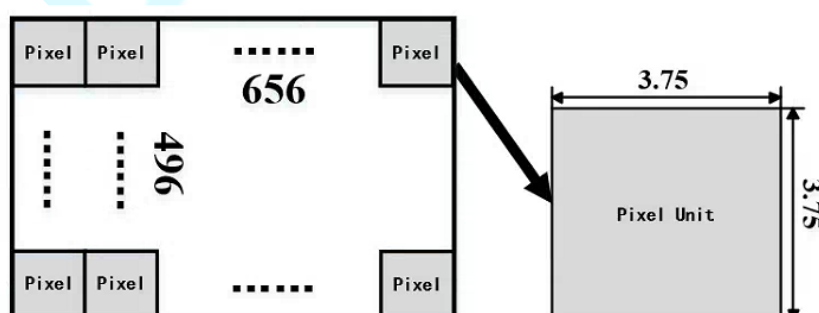
The left side IO definitions of the chip are shown below, where the port numbers are ordered from top to bottom. Coordinate values represent the differences between adjacent coordinates (referring to the spacing between the top edges of two pads). For example, for L2, the coordinate value is the difference between L2 and L1, and so forth (with the top edge of the pad chosen as the unified reference point)

The left side IO definitions of the chip				
Pin Order	Symbol	Description	Note	Coordinates (μ m)
L1	VSS	Digital 1.8V Power Supply	Typical : 1.8V	
L2	VCOM	Common Cathode	Typical : -1V*	126.65
L3	VDD	Digital 1.8V Power Supply	Typical : 1.8V	390.6
L4	VCOM	Common Cathode	Typical : -1V*	390.6
L5	VDD	Digital 1.8V Power Supply	Typical : 1.8V	390.6
L6	VCOM	Common Cathode	Typical : -1V*	390.6
L7	VDD	Digital 1.8V Power Supply	Typical : 1.8V	390.6

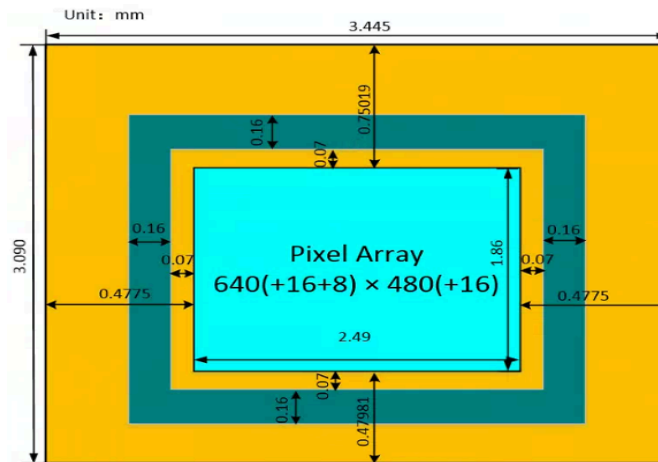
Note:IO ports marked with * require reference to the turn-on voltage of μ LED devices.

4.0 Chip Size

The chip resolution is $640(+16) \times 480(+16)$, with 8 columns of dummy pixels reserved on the right side. These dummy pixels cannot be used for normal display and are only for testing purposes. The pixel unit size is $3.75\mu\text{m} \times 3.75\mu\text{m}$, and the arrangement is shown below. Each pixel unit has an identical structure.



The entire chip dimensions are 3.445mm × 3.090mm. The display area consists of $(640+16) \times (480+16)$ pixels, with the active area (AA) measuring 2.46mm × 1.86mm. The chip dimensions are illustrated below, where the cathode ring (shown as the dark green area in below) has a width of 160μm and a distance of 70μm from the AA area.



5.0 Reliability

Cleaning: It is recommended to use a clean cloth or lens tissue for wiping. Do not use any corrosive chemicals for cleaning.

Electrostatic Protection: The internal chips of the product are made using CMOS technology, which is sensitive to static electricity. It is recommended to use the product in areas away from charged objects or areas with anti-static equipment. Operators should wear appropriate anti-static clothing.

Precautions: It is recommended to use protective gloves when handling the product.

Do not expose the product to environments with corrosive chemicals such as acids or alkalis.

Do not submerge the product in liquid.

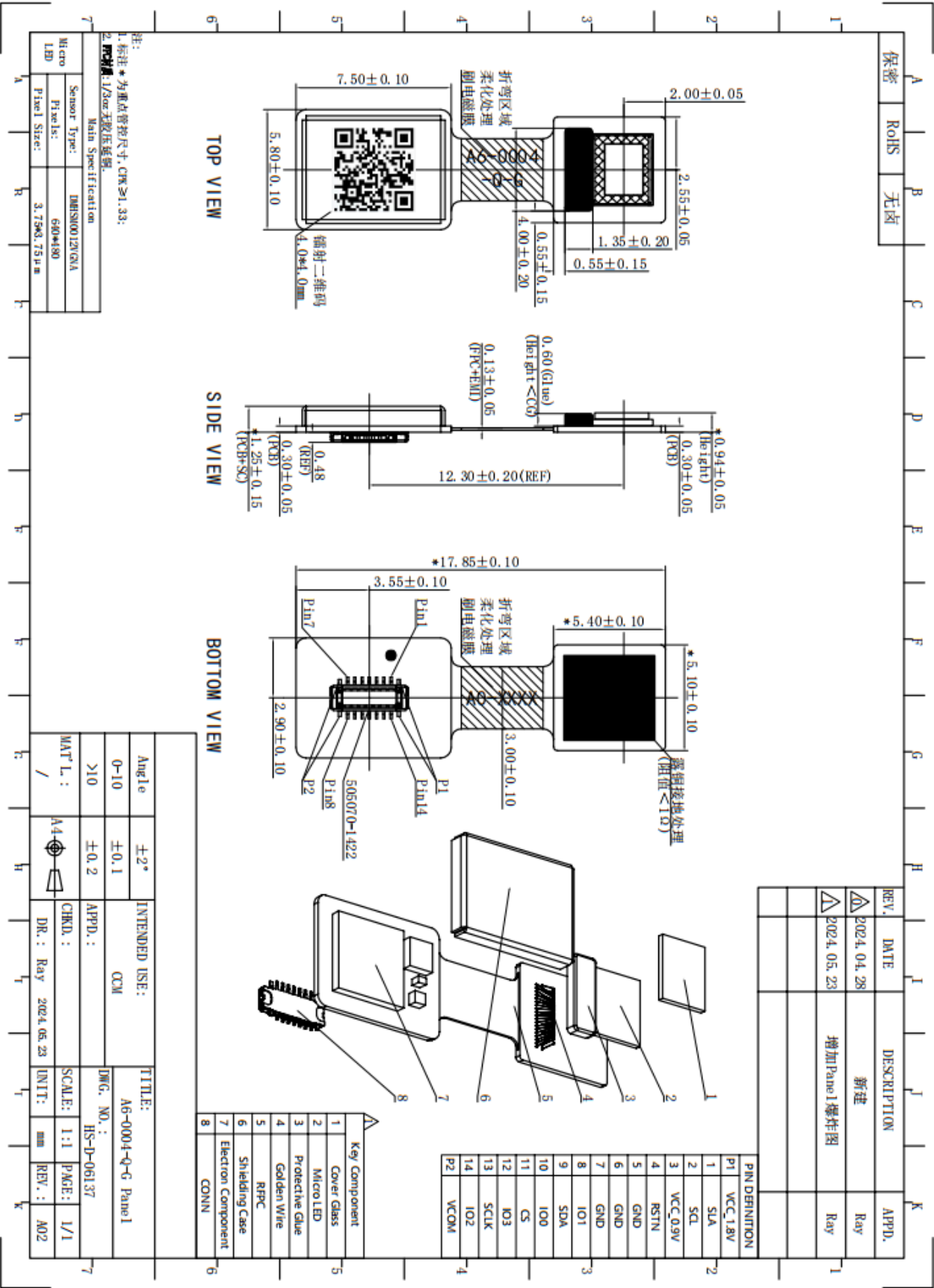
Avoid contact with sharp objects and impacts from external forces.

Storage Method: The product may have a shortened lifespan or be damaged if exposed to excessively high or low temperatures. The recommended storage environment is:

Room temperature: approximately 25°C, fluctuating by $\pm 5^\circ\text{C}$.

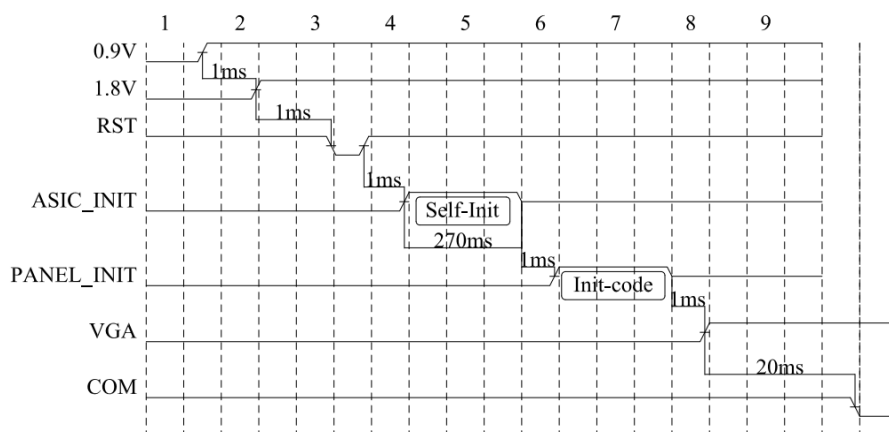
Environment: A dry, sealed container in a stationary state to avoid impacts.

6.0 Mechanical Dimensions(mm)

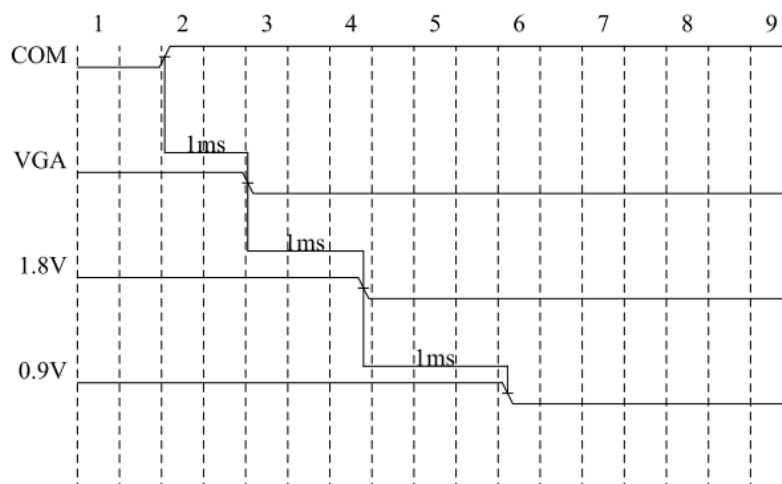


7.0 Power Supply Sequence

7.1 Power up Sequence



7.2 Power down Sequence



8.0 Description of Functions

8.1 SPI/QSPI data communication format description

Support one-line SPI monochrome 8bit display data, support one-line SPI monochrome 6bit display data, support four-Line QSPI monochrome 8bit display data.

Using SPI/QSPI to send display data, the amount of display data sent each time CS pulls down needs to be per row An integer multiple of the amount of data, that is, at least 640 bytes of display data are sent each time CS pulls down.

8.1.1 Monochrome 1 line SPI mode

8.1.1.1 Monochrome 8bit display mode

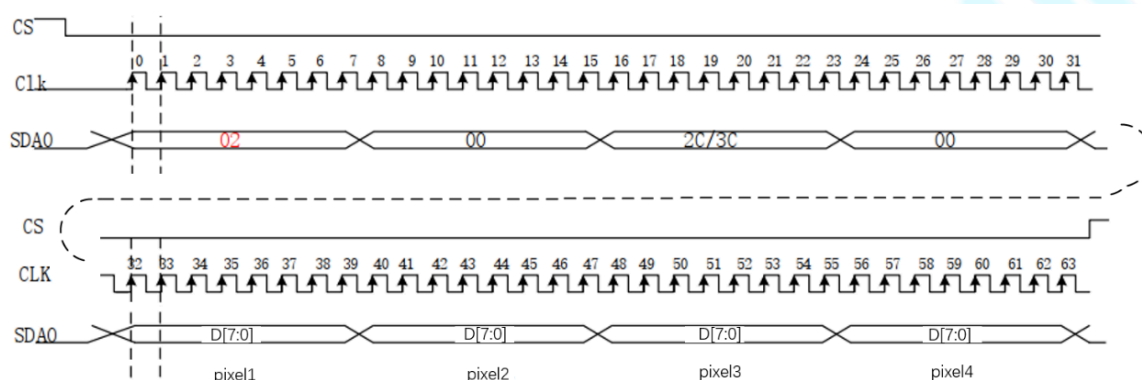


Figure 8-1 Timing diagram of monochrome 8bit1 line SPI display mode

Figure 8-1 shows the sequence of monochrome 8bit 1-line SPI display mode. Send the 8bit command 0x02, and then send the 24-bit command 0x002C00/0x003C00, followed by one pixel of data every 8 bits (send high first Bits). A total of 640*480 pixels of data are sent to end the data transmission of this frame.

8.1.1.2 Monochrome 6bit display mode

The timing is the same as the monochrome 8bit display mode, the only difference is that the 8bit display requires 8 sclk cyclesA pixel data, 6bit display only takes 6 sclk cycles to transmit a pixel data.

8.1.2 4-wire QSPI mode

8.1.2.1 Monochrome GRAY256 display mode

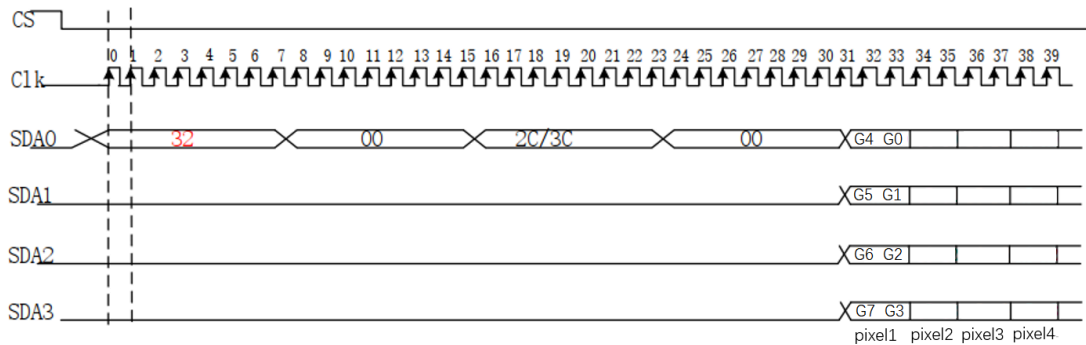


Figure 8-2 Monochrome 4-line QSPI GRAY256 timing diagram

Figure 8-2 shows the monochrome GRAY256 timing sequence. Send the 8bit command 0x32 first, and then send the 24bit command 0x002C00/0x003C00, monochrome GRAY256 sends one pixel in two clock cycles Data.

8.1.3 SPI display partial update

Support for SPI to send partial row display data format, enabling SPI to refresh specific rows of display data.

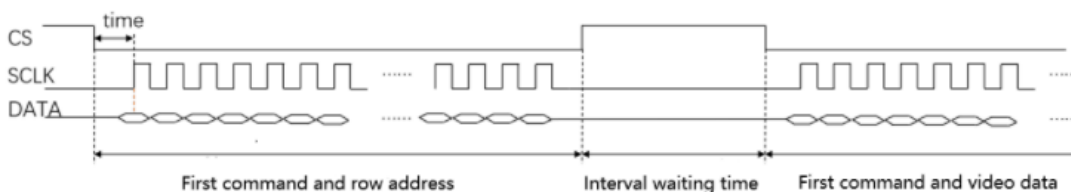


Figure 8-3 SPI partial row format requirements

The SPI partial row format requirements are shown in Figure 8-3. The format is as follows: Send command and row address → Wait for interval time → Send command and display data → Wait for interval time → Send command and row address again. It also supports sending a single row address and then continuously sending multiple display data with the address incrementing internally within the chip: Send command and row address → Wait for interval time → Send command and display data → Wait for interval time → Send display data.

The interval waiting time must be at least 1μs, and the data must be in row units. The time for each operation must be at least 3ns.

8.1.3.1 Command and row address patterns

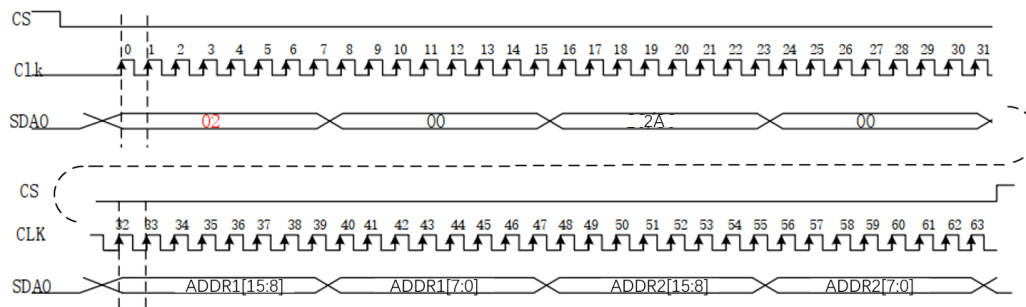


Figure 8-4 SPI sending row address timing

The SPI row address sending timing is shown in Figure 8-4. The sequence is as follows: Send 8-bit command **0x02** → Send 24-bit value **0x002A00** → Send 16-bit **ADDR1** address → Send 16-bit **ADDR2** address.

Here, ADDR1 represents the starting row. For example, in a 640x480 resolution with 480 rows, row addresses range from 0 to 479, where row 0 is the first row, and row 479 is the last row. ADDR2 represents the ending row.

8.2 SPI Configuration Register Description

8.2.1 SPI Configuration Register Function

The SPI configuration allows you to configure 256 8-bit registers. Among these: **SDA0** is used as an input. **SDA1** is used for SPI read output.

8.2.2 SPI Write Register Timing

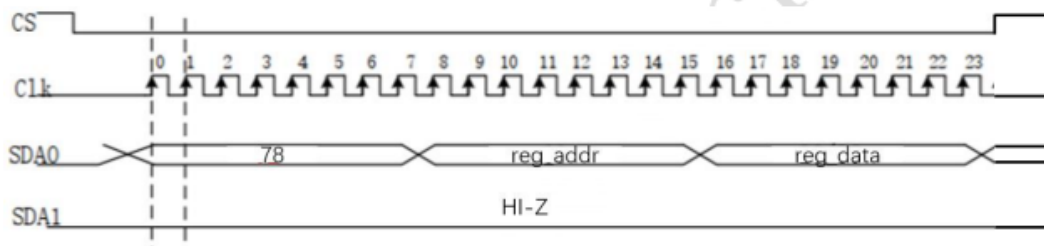


Figure 8-5 SPI Write Register Timing

The SPI write register timing is shown in Figure 8-5. After CS is pulled low, SDA0 sends the command 0x78, then the register address is sent, followed by the register value. SDA1 is not used.

8.2.3 SPI Read Register Timing

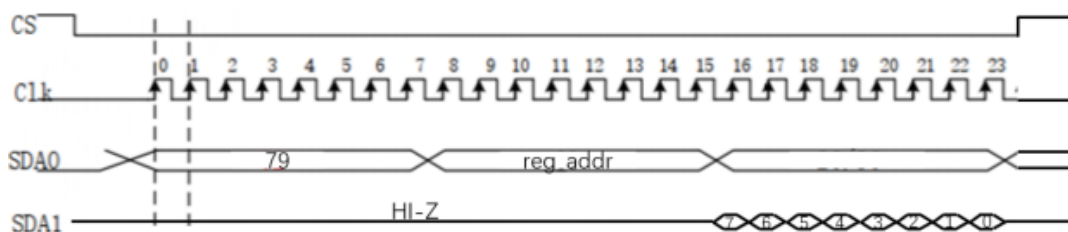


Figure 8-6 SPI Read Register Timing

The SPI read register timing is shown in Figure 8-6. After CS is pulled low, SDA0 sends the command 0x79, then the register address is sent. Next, SDA1 returns the 8-bit register value, with the high byte of the register value sent first.

Continuous read and write operations are not allowed. Only one register can be written (or read) at a time.

8.3 Internal Register Configuration Description

8.3.1 Function Description

- Supports horizontal and vertical mirroring.
- Supports 0-7 level PWM dimming.
- The module supports internal 10-bit GAMMA correction.
- The module internally implements DEMURA correction.
- SPI/QSPI display input data formats:
 - Supports SPI GRAY256, SPI GRAY64, and QSPI GRAY256 display data.
 - The maximum supported speed is 80M.

8.3.2 SPI Register Description

Device address 0x78.

Register address	Bit	Description
0x00	[1:0]	Select display input data format: 00: RGB888, 01: RGB565, 10: GRAY256, 11: GRAY64
	[2]	Keep 0
	[3]	Keep 1
	[4]	Keep 1
	[6:5]	Keep 0
	[7]	Keep 1
0x01	[7:0]	Keep 0
0x02	[7:0]	Keep 0
0x03	[7:0]	Keep 0x95
0x04	[7]	Test Pattern Function 0: Function disabled 1: Function enabled

	[6:5]	Keep 0
	[4:0]	SPI Module Test Pattern Selection: 0: Black screen 1: White screen 2: White screen 3-7: None 8: 2x2 checkerboard 9: 4x4 checkerboard Other values: Not used
0x05-0x15	[7:0]	The 17 gamma segment parameters correspond to the high 8 bits of lut1 to lut17. The default values are: 00 10 20 30 40 50 60 70 80 90 a0 b0 c0 d0 e0 f0 ff These values represent the gamma1.0 parameters, which correspond to the gamma function being turned off.
0x16	[7:0]	bits [1:0], [3:2], [5:4], [7:6] represent the lower 2 bits of Lut2 to Lut5
0x17	[7:0]	bits [1:0], [3:2], [5:4], [7:6] represent the lower 2 bits of Lut6 to Lut9
0x18	[7:0]	bits [1:0], [3:2], [5:4], [7:6] represent the lower 2 bits of Lut10 to Lut13
0x19	[7:0]	bits [1:0], [3:2], [5:4], [7:6] represent the lower 2 bits of Lut14 to Lut7
0x1a	[7:0]	Demura Correction Reference Brightness (ref)
0x1b	[7]	Demura Function and PWM Function: 0: Function disabled. 1: Function enabled. (Note: To disable the Demura function while keeping the PWM function active, set the Demura Correction Reference Brightness (ref) to 0xFF, which will turn off the Demura function.)
	[6:4]	ASIC Self-Test Image: 0: Disable self-test. 1: Two intersecting bright lines along the screen diagonals. 2: Two intersecting dark lines along the screen diagonals. 3: 8 intersecting grayscale bars. 4: 10 vertical grayscale bars incrementing from left to right, with the maximum grayscale determined by the Demura correction reference brightness (ref) value. 5: 10 horizontal grayscale bars incrementing

		from top to bottom, with the maximum grayscale determined by the Demura correction reference brightness (ref) value. 6: Checkerboard pattern. 7: The outermost unclosed bright lines of the screen.
	[3:0]	Keep 0
0x1c	[7]	1: Use the values from the flash address range 0x4F000-0x4F016 to replace the SPI registers 0x05-0x1B. The default is 0.
	[6:5]	Keep 0
	[4]	Vertical Mirroring Function 0: Function disabled 1: Function enabled
	[3]	Horizontal Mirroring Function 0: Function disabled 1: Function enabled
	[2:0]	PWM Dimming 0-7 Level Brightness
0x1d	[7:1]	Keep 0
	[0]	Keep 0
0x1e	[7:0]	Keep 0

9.0 Reference Peripheral Circuits

10.0 Firmware Update

11.0 Certifications

Rohs	✓
CE	×
FCC	×
Other	×

12.0 Packaging Information

10.1 Package Size

10.2 Weight

10.3 ESD package

13.0 Appendix